

PERSONAL CARBON FOOTPRINT CALCULATOR

Know your carbon footprint. Then shrink it.

Answer a few questions about how you travel, use energy, eat and throw things away. CarbonTrace estimates your yearly CO₂ emissions and shows where to cut first.

[Calculate my footprint](#)[How it works](#)**4.7 t**

Approx. world average CO₂ per person per year

1.9 t

Approx. India average per person per year

2.3 t

Commonly cited Paris-aligned level per person by 2030

How it works

Step 1**Enter your habits**

Travel, electricity, LPG, flights, diet and waste, using simple sliders.

Step 2**See your estimate**

Get your annual footprint in tonnes of CO₂ with a breakdown by category.

Step 3**Act on it**

Get a tip aimed at your biggest source and compare with national and global averages.

Why it matters

Carbon dioxide from burning fuel, generating electricity and producing food traps heat in the atmosphere and drives climate change. Individual choices add up across communities, so seeing your own numbers is the first step towards reducing pollution.

[Read practical ways to cut emissions](#)

Your carbon footprint

Main vehicle

Petrol car

Diesel car

CNG car

Electric car

Two-wheeler

Car or two-wheeler (km/week): 50



Bus or train (km/week): 30



Return flights per year: 1



Electricity (kWh/month): 150



LPG cylinders per year: 6



Household waste (kg/week): 7



Diet

Vegetarian

Mixed

Non-vegetarian

Estimated yearly footprint

3.95 t CO₂

Below the world average, but above the 2030 target.

Where it comes from



Compared with



Start here: Home energy is your largest source. Use 5-star appliances and LED lighting, switch off standby loads, and cook with lids on.

Ways to cut your carbon footprint

Each category below matches a part of your calculator result. Start with the one that is largest for you.

Transport

Walk, cycle, use buses or metro, and carpool. Keep tyres inflated and service your vehicle.

Flights

Fly less, choose direct routes, and take trains for shorter journeys.

Electricity

Switch to LEDs and efficient appliances, set the AC to 24-26 °C, and unplug idle chargers.

Cooking gas

Use pressure cookers, cover pots, and plan meals to avoid reheating.

Food

Eat more plant-based meals, buy local and seasonal, and finish what you cook.

Waste

Carry a cloth bag, segregate wet and dry waste, compost, and recycle e-waste properly.

Key terms

Carbon footprint

The total greenhouse gas emissions caused by a person, product or activity, shown as CO₂ equivalent.

Greenhouse gases

Gases such as carbon dioxide and methane that trap heat in the atmosphere.

Emission factor

The amount of CO₂ released per unit of an activity, such as per km driven or per kWh used.

Paris Agreement

The 2015 global agreement to limit warming to well below 2 °C above pre-industrial levels.

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About the project

CarbonTrace is a prototype built for CHE 110 Environmental Studies (CA1). It estimates an individual's yearly carbon footprint from lifestyle choices and suggests where to reduce emissions.

Method and emission factors

Yearly emissions = activity amount × emission factor, summed over five categories. Factors are rounded approximations; real values vary by source and year.

Petrol car	0.17 kg CO ₂ per km
Diesel car	0.15 kg CO ₂ per km
CNG car	0.12 kg CO ₂ per km
Electric car	0.08 kg CO ₂ per km
Two-wheeler	0.05 kg CO ₂ per km
Bus or train	0.04 kg CO ₂ per km
Return flight (average)	250 kg CO ₂ each
Grid electricity	0.71 kg CO ₂ per kWh
LPG cylinder (14.2 kg)	42.5 kg CO ₂ each
Diet (veg / mixed / non-veg)	1000 / 1500 / 1900 kg per year
Household waste	0.45 kg CO ₂ per kg

Technologies

HTML for structure, CSS for styling and JavaScript for the live calculation, all built on one canvas with separate pages for Home, Calculator, Learn and About.

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